

Topic P5: Waves in matter

P5.1 Wave behaviour

Summary

Waves are means of transferring energy and the two main types of wave are introduced in this section: mechanical and electromagnetic. This section considers both what these types of waves are and how they are used. The main terms used to describe waves are defined and exemplified in this topic.

Underlying knowledge and understanding

Learners should have prior knowledge of transverse and longitudinal waves through sound and light. Learners should be aware of how waves behave and how the speed of a wave may change as it passes through different media. They may already have knowledge of how sound is heard and the hearing ranges of different species.

Common misconceptions

Although they will often have heard of the terms ultrasound and sonar, learners find it challenging to explain how images and traces are formed and to apply their understanding to calculations. Learners often misinterpret distance and displacement–time graphical presentations of waves.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
PM5.1i	recall and apply: wave speed (m/s) = frequency (Hz) × wavelength (m)	M1a, M1b, M1c, M2a, M3a, M3b, M3c, M3d

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
P5.1a	describe wave motion in terms of amplitude, wavelength, frequency and period			WS1.1b, WS1.3b, WS1.3e	Observing sound waves on an oscilloscope.
P5.1b	define wavelength and frequency				
P5.1c	describe and apply the relationship between wavelength, frequency and wave velocity		M1a, M1b, M1c, M2a, M3a, M3b, M3c, M3d	WS1.1b, WS1.3a, WS1.3b, WS1.3c, WS1.3d, WS1.3e, WS1.3g, WS1.3h, WS1.3d, WS2a, WS2b	Investigation of reflection in a ripple tank (PAG P4)
P5.1d	apply formulae relating velocity, frequency and wavelength		M1c, M3c		
P5.1e	describe differences between transverse and longitudinal waves	direction of travel and direction of vibration	M5b	WS1.1b, WS1.3e	Use of a slinky to model waves.
P5.1f ☒	show how changes, in velocity, frequency and wavelength, in transmission of sound waves from one medium to another, are inter-related		M1c, M3c		

P5.1g ☒	describe the effects of reflection, transmission, and absorption of waves at material interface	examples such as ultrasound and sonar		WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3e, WS1.3f, WS1.3h, WS2a, WS2b, WS2c	Refraction of light through a glass block. (PAG P8) Investigation of reflection with a plane mirror. (PAG P8) Demonstration of refraction of white light through a prism.
P5.1h ☒	describe, with examples, processes which convert wave disturbances between sound waves and vibrations in solids	knowledge of a simple structure of the parts of the ear is expected		WS1.1b, WS1.1f, WS1.3b, WS1.3e	Use of a signal generator and loudspeaker. Demonstration of sound waves using a Rubens' tube or an oscilloscope.
P5.1i ☒	explain why such processes only work over a limited frequency range, and the relevance of this to human hearing	why hearing (audition) changes due to ageing			
P5.1j	describe how ripples on water surfaces are used to model transverse waves whilst sound waves in air are longitudinal waves, and how the speed of each may be measured			WS1.1b, WS1.3a, WS1.3b, WS1.3c, WS1.3d, WS1.3e, WS1.3g, WS1.3h, WS1.3d, WS2a, WS2b	Investigation of refraction in a ripple tank. (PAG P8)
P5.1k	describe evidence for the cases of ripples on water surfaces and for sound waves in air that it is the wave that travels and not the water or the air				